

# MD SIAM HOSSAIN

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## PROFESSIONAL SUMMARY

Full-Stack Software Engineer with a production-first approach and deep expertise in software architecture, web development, and end-to-end system delivery. Fluent across the entire development lifecycle — backend, frontend, APIs, databases, and deployment. Designs systems with scalability in mind, writes code that's built to maintain, and delivers software that works in the real world, not just on paper. Brings the technical range to own any layer of a product, the engineering discipline to take it from zero to production, and the curiosity to keep improving it after. A fast learner who picks up new technologies, patterns, and domains without losing momentum — and a growth-driven engineer who is never satisfied with yesterday's skill set. Thrives in environments where problems are complex, standards are high, and great software actually matters.

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## EXPERIENCE

### Datasoft Manufacturing and Assembly Inc Limited (DMA)

Jul 2024 – Present

#### Data Science Engineer (Promoted from Intern)

##### BWDB Water Level Anomaly Detection System

- Built an AI-powered anomaly detection system for real-time and historical water level data across 315 BWDB/FFWC river stations.
- Engineered phase-aware dynamic range logic for tidal, non-tidal, and flashy rivers distinguishing HIGH TIDE vs. LOW TIDE behavior, combined with moving averages and adaptive gap-aware thresholds.
- Achieved 97% detection accuracy and improved overall data reliability to 93% through rigorous software testing and validation, directly supporting national flood forecasting and disaster management decisions.

##### System Orchestration Dashboard (SOD)

- Built a centralized dashboard consolidating 8 data engineering tools — gap analysis, database processing, MQTT publishing, file management, interpolation, and hydrological data matching — into one unified interface.
- Developed using React and Django REST Framework with real-time pipeline monitoring.
- Reduced operational complexity by over 70%, cutting pipeline management from multiple manual checks to a single dashboard session.

##### Internal AI Onboarding Chatbot

- Built a RAG-powered chatbot to help employees navigate the virtual office environment independently.
- Implemented using LangChain, FAISS, HuggingFace embeddings, and Groq; integrated query expansion to handle incomplete or ambiguous inputs.
- Actively used by 30+ employees, reducing onboarding query resolution time from hours to seconds.

##### Diagnostic Fault-Detection Agent

- Designing a conversational agent that diagnoses system failures at component level — telling operators exactly what to fix instead of inspecting the entire system.
- Built as an intelligent reasoning layer over live system state data, targeting significant reduction in fault diagnosis time through precise component-level recommendations.

##### IT & Data Engineering

- Performed large-scale data gap filling across 315 station datasets using Cubic Spline and Linear Interpolation methods.
- Managed company servers and automated end-to-end data processing pipelines; performed ongoing software maintenance and code review to ensure code quality, reliability, and zero reported pipeline downtime.
- Ensured consistent data availability across all core operations with zero reported pipeline downtime.

### Office of the Director of Students' Affairs, DIU

Nov 2025 – Present

#### Student Associate

##### DIU Students Hub Platform

- Contributed to full-stack web development, building and enhancing a centralized Laravel web platform serving 30,000+ DIU students across academic, co-curricular, and career needs.
- Built a dynamic Portfolio & CV Generator module with 3 exportable templates, enabling students to generate a professional CV in seconds.
- Developed an Event Management System supporting Central and Club Events with auto-generated brand investment financial proposals.
- Built a Survey Management System with role-based access and graphical reporting dashboards, currently used by 25 departmental teachers.
- Developed a Club Recommendation Agent and a Psychology Agent — two conversational AI tools helping students discover clubs and access mental health support and appointment booking.

### BRRI Agromet Lab, Bangladesh Rice Research Institute

Nov 2025 – April 2026

### Intern Software Developer

#### Agromet Lab Task Tracker

- Built a full-stack custom task and team management system for Dr. Niaz Md. Farhat Rahman, designing the software architecture from scratch to manage projects, meetings, schedules, and collaborations in one place.
- Developed using Django, React, MySQL, and REST API; includes task tracking, intern management, meeting scheduling, and an intelligent KPI-based performance rating engine.
- System actively manages 20+ collaborators, each automatically rated using a weighted composite score across four KPIs — Timeliness (30%), Quality (40%), Efficiency (20%), and Reliability (10%) — with an admin leaderboard and per-collaborator radar chart dashboard.

### IoT Lab, Daffodil International University

Aug 2024 – Jan 2025

#### Student Associate

##### Guardian on Board

- Developed an IoT-powered school bus tracking system using NFC POS machines, GPS, and MQTT for real-time location, on-board status, and departure updates for guardians.
- Achieved 99.6% message delivery success rate; optimised POS battery usage to 35–40% daily consumption.

##### BRRRI Smart AWD Pipe Irrigation System

- Contributed as frontend developer and server/IT administrator to a BRRRI-collaborated smart irrigation system integrating weather forecasting to automate water control in rice fields.
- Optimised irrigation efficiency by up to 40%, reduced water wastage by 35%, and lowered operational costs by 20%.

### IoT Lab, Daffodil International University

Fall 2022 – 2024

#### Regular Member

##### Smart Railway Crossing System (SRCS)

- Built an IoT-based automated railway crossing gate system with self sensor-failure detection and a central control panel for manual override. Published Spring 2023.

##### Individual Water Billing System (IWBS)

- Built a system to monitor individual water consumption per unit and generate bills via a mobile app. Won 1st Runner-up at the 4th International Robotech Olympiad 2024.

### IoT Lab, Daffodil International University

2024 & 2025

#### IoT Trainer

- Trained participants on IoT system design, server management, network and security management, and large-scale data integrity and availability (2025).
- Trained participants on data acquisition, management, and Big Data applications in IoT (2024).

## EDUCATION

### Daffodil International University (DIU)

Jul 2022 – Jul 2026 (Expected)

BSc in Computer Science and Engineering | CGPA: 3.63 / 4.00

### Uttara High School and College (UHSC)

2021

Higher Secondary Certificate — Science | GPA: 5.00 / 5.00

## FINAL YEAR DESIGN PROJECT

### AI-Driven Personalized Learning Path Generation

2025 – 2026

#### Supervisor: Amit Chakraborty Chhoton

- Developing a Deep Knowledge Tracing (DKT) model using LSTM on the EdNet KT1 dataset (829,878 interactions, 6,404 students); achieved 80.3% accuracy and AUC-ROC of 0.9510 in Phase 1.
- Phase 2 implements a Deep Q-Network (DQN) Reinforcement Learning agent for personalized skill-based learning path recommendations.

## PUBLICATIONS

1. M. S. Hossain et al., "Guardian on Board (GoB): IoT-Powered Student Transport Management System," 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025, pp. 1–6. [DOI: 10.1109/QPAIN66474.2025.11172262](https://doi.org/10.1109/QPAIN66474.2025.11172262)
2. A. Sikder, M. S. Hossain et al., "Smart Railway Crossing System (SRCS) Using IoT Philosophy: Bangladesh Context," in *Intelligent Networks and Systems*, Chapman and Hall/CRC, Spring 2023. [View Chapter](#)

INDIVIDUAL PROJECTS

MediAI: Symptom-Based Disease Prediction System

- Built an ANN-based disease prediction tool (TensorFlow/Keras) integrated into a Django web application; achieved 97% prediction accuracy and 0.08% test loss.

Disaster-Aware Urban Planning

- ML framework using Random Forest, Gradient Boosting, and Logistic Regression for flood risk zone classification in Dhaka; achieved 92% accuracy with Gradient Boosting, supporting data-driven infrastructure planning.

QR Code Smart Ticketing System

- IoT + Flutter mobile app as a modern alternative to paper or NFC ticketing.

Hygiene Hub Dashboard

- Hygiene Hub Dashboard — Monitoring dashboard tracking soap and tissue supply status across washrooms in large buildings.

CERTIFICATIONS & TRAINING

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|-------------------------------------------------------------------------------------------------------------------------------|----------|
| Certificate of Internship Completion — Datasoft Manufacturing and Assembly Inc Limited (DMA)                                  | Sep 2024 |
| Certificate of Project Completion — IoT-Driven AWD Irrigation System, BRRI Agromet Lab                                        | 2025     |
| Certificate of Appreciation — IoT Trainer, "Building Industrial IoT: From Hardware to Data Applications", DIU IoT Lab         | 2024     |
| Certificate of Appreciation — IoT Trainer, "Exploring IoT: Introduction to Hardware, Firmware, and Data Systems", DIU IoT Lab | 2025     |
| Certificate of Presentation — IEEE QPAIN 2025, Guardian on Board (GoB)                                                        | Aug 2025 |
| Certificate of Appreciation — BIM 2023, Smart Railway Crossing System (SRCS)                                                  | Sep 2023 |

TECHNICAL SKILLS

|                         |                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| Machine Learning & AI   | TensorFlow, Keras, Scikit-Learn, Regression, Classification, Clustering, Neural Networks, Deep Learning |
| AI Agents & NLP         | LangChain, FAISS, HuggingFace, Groq, RAG Pipelines, BanglaBERT, XLM-RoBERTa, Named Entity Recognition   |
| Data Analysis           | Pandas, NumPy, SQL, Matplotlib, Seaborn, Tableau, Statistical Analysis                                  |
| Backend Development     | Django, Laravel, Node.js, REST API Design, MySQL, PostgreSQL                                            |
| Frontend & Mobile       | React, React Native, Flutter                                                                            |
| IoT                     | Arduino, Raspberry Pi, MQTT, LoRa, Real-time Sensor Integration, Cloud Integration                      |
| DevOps & Infrastructure | Git, GitHub, Docker, Linux Server Administration, Nginx, Gunicorn, Pipeline Automation                  |
| Languages               | Bangla (Native), English (Fluent)                                                                       |

AWARDS & ACTIVITIES

- 1st Runner-up | 4th International Robotech Olympiad 2024 — Individual Water Billing System (IWBS)
- 1st Runner-up | BIM-2023 Project Showcasing Competition — IoT-Based Smart Railway Crossing System
- Section First | TakeOff Programming Contest, Fall 2022 — DIU
- Organizer | DIU IoT Fest 2023 — Organized the first-ever IoT Fest in Bangladesh, bringing together industry experts, students, and innovators